

ICAD RT-4 · Set 7 — Answer Sheet

Exam 12-10-2026 · 75 Q · 300 marks · 3 hours · +4 correct, -1 wrong, 0 blank

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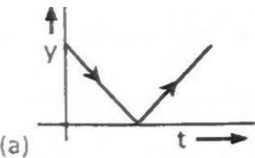
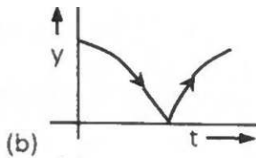
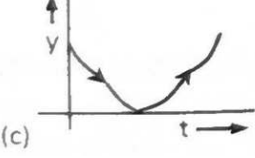
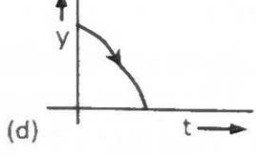
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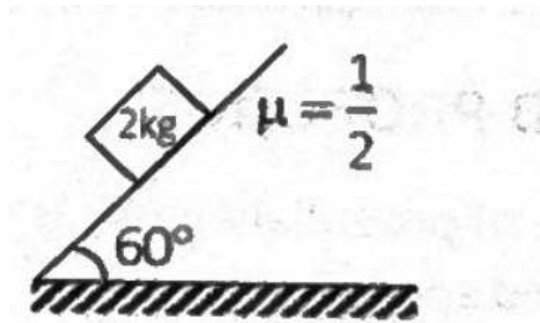
Bubbles — single correct · Box — write the value · each subject out of 100

Physics

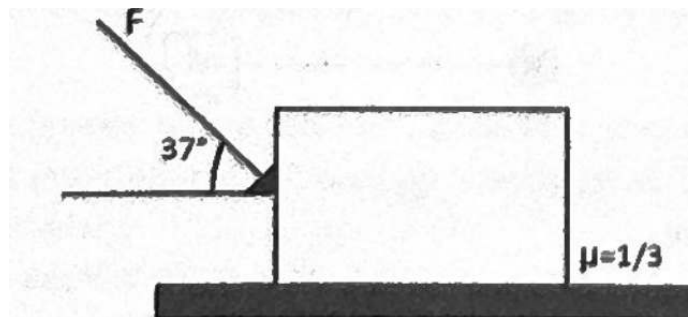
PART A · Q1–20 · SINGLE CORRECT

1. P1L1V4Q2 If $\tan(\theta - \pi/4) = 1$, then $\theta = ?$ (a) $\pi/2$ (b) $\pi/8$ (c) π (d) none
 2. P2L4V5Q9 Claim: "If $y''(c) < 0$, $y(c)$ is a local maximum." Best evaluation? (False unless $y'(c) = 0$). (a) Always true. (b) False unless $y'(c) = 0$. (c) True only if $y(c) > 0$. (d) False unless $y''(x) < 0$ for all x
 3. P2L6V3Q5 If $F(r) = -k/r^2$, find $U(r) = -\int F(r)dr$ under the limit ∞ to r . (a) k/r (b) $-k/r$ (c) k/r^2 (d) $-k/r^2$
 4. P3L2V5Q4 (Multiple Correct) If three vectors are added, the resultant: (a) may be zero (b) must be zero (c) may be non-zero (d) must be non-zero
 5. P3L5V2Q5 The angle between the vectors $a=3\hat{i}-4\hat{j}$ and $b=\hat{i}+2\hat{j}$ is: ($\cos^{-1}(-1/\sqrt{5})$). (a) $\cos^{-1}(-1/\sqrt{5})$ (b) $\cos^{-1}(-1/4)$ (c) $\cos^{-1}(-1/\sqrt{2})$ (d) $\cos^{-1}(-1/\sqrt{6})$
 6. P4L4V4Q4 A ball is dropped from a bridge at a height of 176.4 m over a river. After 2 s, a second ball is thrown straight downwards. What should be the initial velocity of the second ball so that both hit the water simultaneously? (a) 2.45 (b) 49 (c) 14.5 (d) 24.5 m/s
 7. P4L5V5Q3 A particle moves along OX: $x = 40 + 12t - t^3$. How far would the particle travel before coming to rest? (a) 16 m (b) 24 m (c) 40 m (d) 56 m
 8. P4L6V6Q5 A ball is dropped on a floor and bounces back to a height somewhat less than the original height. Which of the curves depicts its motion correctly?
- 



9. P4L10V2Q2 A child runs at 9 km/h (w.r.t. the belt) between parents 50 m apart on a belt moving at 4 km/h. Speed of the child in the direction of the belt's motion, as seen from the platform? (a) 9 km/h (b) 4 km/h (c) 13 km/h (d) 5 km/h
 10. P5L3V2Q4 A force of 12 N is acting at a distance of 0.5 m. What is the torque? (a) 4 Nm (b) 6 Nm (c) 8 Nm (d) 10 Nm
 11. P6L5V2Q4 Bird in a wire cage hanging from a spring balance reads W_1 ; if the bird flies inside, reading W_2 . $W_1=W_2$ / $W_1>W_2$ / $W_1<W_2$ / Nothing definite
 12. P6L7V5Q8 Balloon + contents (M) move up at a ; mass to release so it moves up at $3a$ ($2Ma/(3a+g)$ / $3Ma/(2a-g)$ / $3Ma/(2a+g)$ / $2Ma/(3a-g)$)
 13. P7L3V5SuggQ1 At angle of repose α , which of the following is true? (a) Limiting friction = 0 (b) Driving force = 0 (c) Limiting friction = Driving force (d) Frictional force > Driving force
 14. P8L1V4Q7 A bicycle chain of length 1.6 m and of mass 1 kg is lying on a horizontal floor. If $g = 10 \text{ m s}^{-2}$, the work done in lifting it with one end touching the floor and the other end 1.6 m above the floor is: (a) 10 J (b) 3.2 J (c) 8 J (d) 16 J

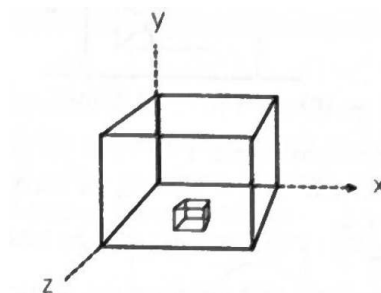
15. P8L2V3Q4 A spring of spring constant $5 \times 10^3 \text{ N/m}$ is stretched initially by 5 cm from the unstretched position. The work required to further stretch the spring by another 5 cm is (a) 6.25 N-m (b) 12.50 N-m (c) 18.75 N-m (d) 25.00 N-m
16. P8L2V5Q8 A force $F = (3x^2y\hat{i} + x^3\hat{j}) \text{ N}$ is acting on a particle. Find work done by the force when the particle moves from (1 m, 1 m) to (2 m, -1 m). (a) 2 J (b) -5 J (c) 8 J (d) -9 J
17. P8L6V3Q3 A block of mass 2 kg is released from rest on a rough inclined ground of inclination 60° with $\mu = 1/2$, as shown in the figure. Find the work done on the block by (a) Gravity, (b) Force of friction, when the block is displaced downwards along the plane by 2 m. (Take $g = 10 \text{ m/s}^2$.)



18. P8L1V4Q8 A block of mass $M = 9 \text{ kg}$ is moving with uniform velocity of 2 m/s on a rough horizontal surface ($\mu = 1/3$) under the action of a constant force F . The force acts at an angle 37° to the horizontal (pressing down onto the block, as in the figure). Find the work done by the force F during an interval of 4 s of motion. (a) 400 J (b) 300 J (c) 320 J (d) 100 J



19. P6L7V5Q10 2 kg block at rest on a box with velocity $2\hat{i}$ and acceleration $-3\hat{i} + 4\hat{j} \text{ m/s}^2$ — pseudo force on the block w.r.t. box (write the value)

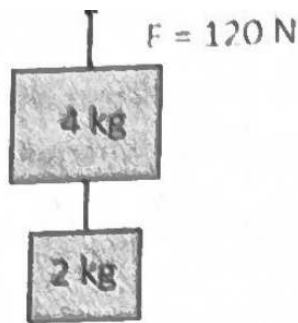


20. P7L1VBoardQ3 A 20 kg metal block is placed on a horizontal surface. The block just begins to slide when a horizontal force of 100 N is applied to it. Calculate the coefficient of static friction. If the coefficient of kinetic friction is 0.4, find the minimum force needed to maintain its uniform motion. ($g = 9.8 \text{ m/s}^2$) (write the value)

PART B · Q21–25 · NUMERICAL VALUE

21. P5L3V4Q4 A body of mass 10 kg is suspended by a rope 50 cm long and pulled to a side through a horizontal distance of 30 cm by a horizontal force. Find that horizontal force (newton)

22. P6L7V4Q4 4 kg & 2 kg joined by a light string, pulled up by $F = 120$ N — acceleration of the blocks (m/s^2)



23. P3L6L7V5Q8 A ray of light travelling along $\frac{1}{2}(\hat{i} + \sqrt{3}\hat{j})$ is incident on a plane mirror; after reflection it travels along $\frac{1}{2}(\hat{i} - \sqrt{3}\hat{j})$. Mark the correct option(s) about the angle of incidence/reflection

Chemistry

PART A · Q1–20 · SINGLE CORRECT

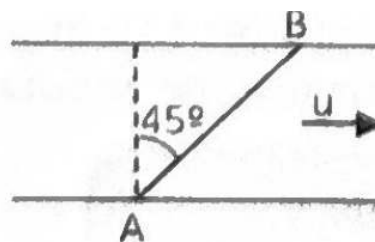
1. C2L3V2Q3 Stopping potential $V_0 = 3.0$ V — calculate max KE of emitted photoelectrons. (a) 4.8×10^{-19} J (b) 3.0×10^{-19} J (c) 1.6×10^{-19} J (d) 9.6×10^{-19} J
2. C2L5V6Q6 Binding energy 13.6 eV — energies to remove electron from first three excited states. (a) 1.51, 0.85, 0.54 eV (b) 6.8, 3.4, 1.51 eV (c) 13.6, 6.8, 3.4 eV (d) 3.4, 1.51, 0.85 eV
3. C2L8V3Q1 Electron $\Delta x = 1.0 \times 10^{-10}$ m — minimum uncertainty in momentum Δp . (a) 5.28×10^{-25} kg.m.s⁻¹ (b) 1.06×10^{-24} kg.m.s⁻¹ (c) 3.31×10^{-26} kg.m.s⁻¹ (d) 6.63×10^{-25} kg.m.s⁻¹
4. C2L11V4Q2 Electronic configuration of Pd^{2+} (at. no. 46). (a) $[\text{Kr}]4d8$ (b) $[\text{Kr}]5s2 4d6$ (c) $[\text{Kr}]4d6$ (d) $[\text{Kr}]4d8 5s2$
5. C3L7V3Q3 18 g sugar in 150 mL solution. Calculate W/V% and sugar in 100 mL. (a) 12%, 12 g (b) 9%, 9 g (c) 15%, 10 g (d) 18%, 18 g
6. C3L7V4Q7 10 g of glucose is dissolved in 150 g of water. The mass percentage of glucose is: (a) 2.50% (b) 6.25% (c) 8.75% (d) 10%
7. C3L8V4Q7 1 kg sea water contains 6 mg dissolved O_2 ; concentration in ppm. (a) 0.6 (b) 6.0 (c) 60.0 (d) 16.0
8. C3L9V4Q6 100 mL each of 0.5 N NaOH, N/5 HCl and N/10 H_2SO_4 mixed — nature? (a) Acidic (b) Neutral (c) Alkaline (d) None
9. C4L11V5Q10 As we go from left to right in the 2nd period of the periodic table, the gram atomic volume of the elements: (a) first decreases then increases (b) decreases (c) remains unchanged (d) will change indefinitely
10. C4L3V3Q5 Atomic radius of Al = 143 pm. Radius of Al^{3+} = 53 pm. Difference? (a) 90 pm (b) 100 pm (c) 80 pm (d) 70 pm
11. C5L1V6Q2 Amongst LiCl, RbCl, BeCl_2 and MgCl_2 , the compounds with the greatest and the least ionic character respectively are: (a) LiCl and RbCl (b) RbCl and BeCl_2 (c) RbCl and MgCl_2 (d) MgCl_2 and BeCl_2

PART B · Q21–25 · NUMERICAL VALUE

21. C1L11V3Q1 Dipole interaction $\propto 1/r^3$ (solids), 2 Å apart — how much stronger at 1 Å? (a) 8 (b) 4 (c) 2 (d) 16

24. P4L7V5Q5 A stone aimed at point P (same horizontal plane) falls x_1 short at 15° , lands x_2 beyond at 45° ; correct angle to hit P is 30° . Find $(\sqrt{3}-1)x_1/x_2$

25. P4L12V4Q1 A man wants to reach point B on the opposite bank of a river flowing at speed u , along a path at 45° as shown. What minimum speed relative to water should he have, if $u = 5\sqrt{2}$?



12. C5L2V4Q1 What are the exceptions of the octet rule? (a) The incomplete octet of the central atom (b) An odd number of electrons on the central atom (c) Expanded octet of the central atom (d) All of these
13. C5L3V5Q6 The correct sequence of bond length in single bond, double bond and triple bond of C is (a) $(\text{C}-\text{C}) = (\text{C}=\text{C}) = (\text{C}\equiv\text{C})$ (b) $\text{C}\equiv\text{C} < \text{C}=\text{C} < \text{C}-\text{C}$ (c) $\text{C}-\text{C} < \text{C}=\text{C} < \text{C}\equiv\text{C}$ (d) $\text{C}=\text{C} < \text{C}\equiv\text{C} < \text{C}-\text{C}$
14. C5L9V3Q3 Although N–F bonds are more polar than N–H bonds, the dipole moment of NF_3 is smaller than that of NH_3 because: (a) NF_3 is heavier than NH_3 (b) Lone-pair moment adds in NH_3 but partly cancels in NF_3 (c) Bond length of N–F is shorter (d) Fluorine has lower electronegativity than hydrogen
15. C5L9V3Q4 Which conclusion can be drawn if a molecule containing multiple polar bonds has zero dipole moment? (a) The molecule is non-polar due to zero electronegativity difference (b) The molecule is symmetrical in shape (c) The molecule has no lone pairs (d) The molecule must be diatomic
16. C5L9V5Q2 Among the following, the paramagnetic compound is: (a) Na_2O_2 (b) O_3 (c) N_2O (d) KO_2
17. C3L1V5Q9 3.2×10^5 atoms weigh 8.0×10^{-18} g; atomic mass. (a) 2.5×10^{-22} (b) 15 (c) 8.0×10^{-18} (d) 30
18. C5L6V3Q1 Determine hybridisation and bond angle in CO_2 . (a) sp, 180° (b) sp^2 , 120° (c) sp^3 , 109.5° (d) sp, 120°
19. C1L7V6Q10 Assertion: At a fixed temperature, the difference between the RMS speed and the average speed of gas molecules decreases as the molar mass of the gas increases. Reason: Both RMS speed and average speed of gas molecules are directly proportional to the square root of temperature and inversely proportional to the square root of molar mass. (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true, but R is false. (d) A is false, but R is true.
20. C1L9V3Q5 Given $a=3.6$, $b=0.0427$ — calculate P c using $P c = a/(27b^2)$. (a) 42.5 atm (b) 77.3 atm (c) 44.1 atm (d) 73.1 atm

22. C1L5V4Q3 A ($M=25$); B takes 3× longer — molar mass $M(B)$. (a) 225 (b) 100 (c) 75 (d) 50

- 23.** C2L11V4Q1 Atom has $2e^-$ in K, 8 in L, 8 in M — number of p-electrons?
- 24.** C5L4V5Q7 Among B_2H_6 , $B_3N_3H_6$, N_2O , N_2O_4 , $H_2S_2O_8$, H_2SO_5 — the total number of molecule containing covalent bond between two atoms of the same kind is _____

Mathematics

PART A · Q1–20 · SINGLE CORRECT

- 1.** M1L2V6Q1 At 6:15, difference between the two angles made by the hands. (a) 165° (b) 170° (c) 175° (d) 180°
- 2.** M2L3V3Q3 $A = \{2, 3, 4, 8, 10\}$, $B = \{3, 4, 5, 10, 12\}$, $C = \{4, 5, 6, 12, 14\} \rightarrow (A \cup B) \cap (A \cup C)$. (a) $\{2, 3, 4, 5, 8, 10, 12\}$ (b) $\{2, 4, 8, 10, 12\}$ (c) $\{3, 8, 10, 12\}$ (d) $\{2, 8, 10\}$
- 3.** M2L3V6Q2 $A, B, C \subseteq U$; pairwise $\cap$ non-empty but $A \cap B \cap C = \emptyset \rightarrow$ which sets qualify (multi-correct). (a) $A = \{1, 2\}$, $B = \{2, 3\}$, $C = \{1, 3\}$ (b) $A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$, $C = \{5, 6, 7\}$ (c) $A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$, $C = \{2, 3, 5\}$ (d) $A = \{a, b\}$, $B = \{b, c\}$, $C = \{a, c\}$
- 4.** M3L1V4Q5 $5 \sin X + 4 \cos X = 3 \rightarrow 4 \sin X - 5 \cos X$. (a) 4 (b) $4\sqrt{2}$ (c) $3\sqrt{2}$ (d) $\sqrt{2}$
- 5.** M3L3V5Q10 Assertion $\sin 1 < \sin 1^\circ$; Reason $\sin$ decreasing on $(0, \pi/2)$. (a) A, R are true and R is the correct explanation of A (b) A, R are true and R is not the correct explanation of A (c) Both A & R are false (d) A is true, R is false
- 6.** M4L1V5Q4 $2^x = 2^{((x+y)/5)} \rightarrow 2^{((y/x)-3)}$. (a) 2 (b) $1/2$ (c) -2 (d) none
- 7.** M4L4V2Q2 If $y = \sin(x + \pi/2)$, then y equals (a) $-\cos(x)$ (b) $\sin(x)$ (c) $\cos(x)$ (d) $-\sin(x)$
- 8.** M5BL1V4Q9 If $3 \sin \theta + 4 \cos \theta = 5$, then $4 \sin \theta - 3 \cos \theta = ?$ (a) 0 (b) 5 (c) 1 (d) none
- 9.** M5CL4V4Q4 $A + B + C = \pi \Rightarrow \tan A + \tan B + \tan C =$ (a) $\sin A \sin B \sin C$ (b) $\cos A \cos B \cos C$ (c) $\tan A \tan B \tan C$ (d) $\cot A \cot B \cot C$
- 10.** M6L5V5Q6 If $(5 - 4x)/(3x^2 - x - 4) < 0$, then: (a) $(-\infty, -1) \cup (5/4, 4/3)$ (b) $(-1, 5/4) \cup (4/3, \infty)$ (c) $(-1, 4/3)$ (d) none of these
- 11.** M7L1V5Q5 If $2^{(1-x)} = (\sqrt{2})^{(x/2)}$, then the value of x is: (a) $4/5$ (b) $5/4$ (c) $-1/3$ (d) none

PART B · Q21–25 · NUMERICAL VALUE

- 21.** M5BL2V3Q4 Find the value of $(\sin 15^\circ - \cos 15^\circ)^2$.
- 22.** M6L4V3Q2 If solution set of $(5 - 2x)/3 \leq x/6 - 5$ is $[a, \infty)$, then value of $(a + 9)$ is _____
- 23.** M3L1V6Q2 $10 \sin^4 \alpha + 15 \cos^4 \alpha = 6 \rightarrow 27 \operatorname{cosec}^6 \alpha + 8 \sec^6 \alpha$

- 25.** C1L10V3Q2 1.5 N force over length 5 cm — surface tension in N/m. (a) 30 (b) 25 (c) 20 (d) 15

- 12.** M7L2V6Q5 For a in $(0, 1)$, which statement about $f(x) = \log_a(x)$ and $g(x) = a^x$ is true? (a) Both are increasing and intersect exactly once (b) f is decreasing, g is decreasing; they never intersect (c) f is decreasing, g is decreasing; they intersect at most once (d) f is decreasing, g is increasing; they can intersect at most once
- 13.** M8AL2V5Q7 If the sum of n terms of an A.P., is $3n^2 + 5n$ then which of its terms is 164? (a) 26 th (b) 27 th (c) 28 th (d) none
- 14.** M8BL3V3Q4 If the n th term of an A.P. be $(2n - 1)$, then the sum of its first n terms will be (a) $n^2 - 1$ (b) $(2n - 1)^2$ (c) n^2 (d) $n^2 + 1$
- 15.** M8BL3V3Q5 If the sum of three terms of G.P. is 19 and product is 216, then the common ratio of series is (a) $-3/2$ (b) $3/2$ (c) 2 (d) 3
- 16.** M8BL3V5Q2 In a G.P. of even number of terms, the sum of all terms is five times the sum of the odd terms. The common ratio of the G.P. is (a) $-4/5$ (b) $1/5$ (c) 4 (d) none
- 17.** M8BL3V5Q4 The sum of first two terms of a G.P. is 12. The sum of 3 rd & 4 th term is 48. If the terms of the G.P. are alternately positive & negative, then the first term is (a) 12 (b) 4 (c) -4 (d) -12
- 18.** M8CL1V4Q9 If $(a + be^y)/(a - be^y) = (b + ce^y)/(b - ce^y) = (c + de^y)/(c - de^y)$, then a, b, c, d are in (a) A.P. (b) G.P. (c) H.P. (d) A.G.P.
- 19.** M4L2V5Q7 Degree-5 poly, leading unity, $f(1)=5 \dots f(5)=1 \rightarrow f(6)$. (a) 0 (b) 24 (c) 120 (d) 720
- 20.** M7L3V5Q4 If $(\log_2 a)/2 = (\log_3 b)/3 = (\log_4 c)/4$ and $a^{(1/2)} b^{(1/3)} c^{(1/4)} = 24$, then: (a) $a = 24$ (b) $b = 81$ (c) $c = 64$ (d) $c = 256$

- 24.** M5AL3V5Q1 Find the maximum value of $A = \sin^2 \theta + \cos^4 \theta$.

- 25.** M6L3V4Q7 If $|x + 1/3| > 8/3$, then find the number of integral values which do NOT satisfy the inequality.